

Identifying Market Gaps for High-End Grocery Stores in Los Angeles

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Executive Summary

Selecting locations for a new store is one of the most important decisions that high-end grocery chains/stores face, where a ‘bad’ location can lock in years of underperformance while a strong location can focus on both revenue and brand identity. This paper focuses on where three premium grocery stores such as Erewhon, Whole Foods, and Bristol Farms should expand in the Los Angeles area, where current market gaps exist, and how store placement can relate to neighborhood affluence and demographics. Los Angeles is a useful setting/location because of its diverse variation of incomes, lifestyle, and density across different zip codes, which provides the cross-sectional variation needed to estimate location effects credibility.

To build our analysis, we focused on four main sources from existing literature. Hotelling’s theory of spatial competition helps explain why premium grocers tend to group together and offer similar products instead of differentiating strongly on price. Huff’s probabilistic gravity model gives us a way to think about trade areas as overlapping zones shaped by store size and travel time rather than fixed boundaries. Reed et al. provide empirical evidence on the demographic and lifestyle factors that predict where specialty grocers actually choose to locate, such as income, education, and the presence of fitness establishments. Finally, Smith et al. show that demand for organic and premium food is strong among urban, affluent, and Western households, especially those with young children. Taken together, these works suggest that market gaps for high-end grocery retail should be identifiable in Los Angeles neighborhoods that meet the demand conditions but currently lack a store.

For our data, we put together a Zip code-level dataset by combining store counts from each chain’s official website with neighborhood variables from Census Reporter, Zillow, and City of Los Angeles sources. Our independent variables include median household income, home prices, median gross rent, population density, median age, a walkability index, parking availability, and a competitor dummy for nearby chains like Trader Joe’s, Gelson’s, and Sprouts. Because our dependent variable is a count with a lot of zeros and likely overdispersion, we use a Negative Binomial regression estimated by maximum likelihood. Before estimating the model, we check for multicollinearity using a correlation matrix and Variance Inflation Factors, and either drop or combine variables that are too highly correlated.

The goal of the paper is to figure out which neighborhood characteristics drive where premium grocers locate and to identify high-affluence ZIP codes that have fewer stores than the model predicts. These ZIP codes are our main candidates for expansion, and they help highlight where demand for high-end grocery store options in Los Angeles may currently be underserved.

1. Background

The specialty grocery industry has grown substantially over the past two decades, driven by rising consumer interest in organic produce, artisanal goods, and health trends. Retailers like Erewhon, Whole Foods, and Bristol Farms have emerged as defining players in this space, positioning themselves not merely as food stores but as lifestyle brands with loyal, affluent customer bases. As these chains look to grow, site selection has become one of the most consequential strategic decisions they face. Unlike mass-market grocers that can absorb a poorly performing location within a large portfolio, high-end specialty grocers depend heavily on demographic alignment; their business model requires a critical mass of customers willing to pay a significant premium for the shopping experience they offer. A single misplaced store can mean years of underperformance and reputational risk in a market where brand positioning is paramount.

This analysis focuses on Los Angeles, a city that offers an unusually rich space for studying high-end retail geography. LA combines extreme income inequality, dense and highly segmented neighborhoods, and a large existing base of premium grocery activity making it ideal for examining both where the market has already taken hold and where latent demand may remain unmet. The city's sprawling geography also means that access to high-end grocery options is far from uniform: residents of some neighborhoods enjoy multiple nearby grocery options while others, potentially with comparable spending power, have none.

A central question motivating this work is the relationship between store placement and neighborhood affluence. While it is well established that high-end grocery chains tend to cluster in wealthier areas, the nature of that relationship is less clear. These retailers may simply follow existing wealth, targeting neighborhoods that

already exhibit the demographic profile their business model requires; or, their presence may itself contribute to neighborhood change over time by attracting higher-income residents and accelerating gentrification. Disentangling correlation from causation is beyond the scope of any single study, but understanding the strength and structure of the relationship between income and store placement is a necessary first step toward either interpretation.

2. Literature Review

The question of where specialty grocers choose to locate sits at the intersection of retail location theory, consumer demand economics, and urban demography. For Los Angeles, a city defined by socioeconomic stratification, a health-conscious consumer based, and dense retail competition, existing literature provides a rigorous foundation for evaluating where an unmet gap for high-end grocery retail exists.

In his landmark 1929 paper “Stability in Competition,” Harold Hotelling provides the theoretical foundation for understanding competitive dynamics in retail markets, which applies readily to the study of high-end grocery competition in Los Angeles. Hotelling’s central insight is that prior competition theorists, namely Cournot, Bertrand, and Edgeworth, erred in assuming that consumers always patronize the cheapest available seller. This assumption that infinitesimal increases in prices result in total loss of customers produces unstable price dynamics inconsistent with real market behavior (Hotelling 44). Instead, Hotelling demonstrates that consumers sort themselves among competing sellers based on a combination of price and what he calls “transportation cost” (Hotelling 41). Hotelling uses transportation costs as a catch-all to describe any friction that creates loyalty or switching costs, including but not limited to physical distance, habit, social ties, and preference. When one seller raises prices modestly, customers shift away gradually rather than all at one

because many remain attached to that seller for reasons unrelated to price. This means that each seller effectively acts as a local monopolist over a captive segment of customers, and that a stable price equilibrium exists from which neither competitor has incentive to deviate (Hotelling 43).

Hotelling expands this spatial logic beyond geography into the broader domain of product differentiation. He posits that competing sellers face an incentive to create products just different enough from one another as to capture consumers from the middle of the market while avoiding a price war (Hotelling 54). This tendency toward excessive similarity, which Hotelling observes across industries from cider merchants to political parties, maps directly onto the observable convergence among high-end grocers in Los Angeles (Hotelling 54). Stores like Erewhon, Bristol Farms, and Whole Foods occupy adjacent positions in the quality-and-lifestyle segment of the grocery market, differentiating themselves through subtle variations in sourcing philosophy, private label offerings, and store ambiance rather than through sharp contrasts in price or product category. Hotelling's model predicts precisely this outcome: new entrants into a competitive market have strong incentives to position themselves near established rivals to capture the large middle of health- and quality-conscious consumers rather than staking out a distant and thinly populated niche.

Where Hotelling provides the theoretical logic of spatial competition, David Huff's 1963 probabilistic gravity model supplies the operational tools to measure it. Huff defines a retail trade area as the geographic region containing the probable consumers for a good or service, and argues that the likelihood of a consumer patronizing a given store is a function of two variables: the square footage of the store, used to approximate the variety of goods available, and the travel time required to travel from the consumer's home to the store (Huff 83). In this sense, Huff

translates Hotelling's abstract notion of transportation cost into an empirically measurable quantity, grounding the theoretical framework in observable data. Huff finds that a shopping center's retail trade area can be estimated by dividing the surrounding area into small sections, calculating the probability of consumers from each section patronizing a particular center, and drawing contour lines connecting sections with equal probabilities (Huff 86). Rather than a hard boundary enclosing a shopping center, the trade area is better understood as a series of radial zonal probability contours that fade gradually with distance and overlap with those of competing stores (Huff 87). Huff further proposes that the basic model can be extended with additional data to estimate expected shopping trips over a given time period (Huff 90). The model has become foundational to retail location theory precisely because it is empirically grounded, flexible across product categories, and relatively inexpensive to implement.

Building on his foundational model, Huff's 1964 paper makes an additional methodological contribution by demonstrating the limitations of the dominant prior approaches, particularly Reilly's Law of Retail Gravitation and Converse's breaking-point formula, and showing how the probabilistic framework resolves them. The Reilly-Converse tradition treated trading area boundaries as hard lines, assigning consumers deterministically to whichever competing center dominated at a given point. Huff identifies three core problems with this approach: it cannot produce graduated demand estimates above or below the break-even position between two centers; it generates overlapping or unaccounted-for areas when applied to multi-store geographies; and it incorrectly treats the distance-decay parameter as a universal constant across all shopping trip types (Huff 36). Huff's probabilistic model addresses each of these shortcomings by producing continuous demand surfaces rather than

fixed boundaries, accounting for all competing destinations simultaneously, and allowing the travel-sensitivity parameter to vary by product category, reflecting the intuitive reality that consumers are willing to travel further for specialty purchases than for routine groceries (Huff 37).

Critically for the study of high-end grocery retail in Los Angeles, Huff demonstrates that the distance-sensitivity parameter, how steeply patronage probability declines with travel time, is not a universal constant but varies by product category and shopping trip type. Specialty and high-involvement purchases tend to have lower distance sensitivity, meaning consumers are willing to travel further, while routine commodity purchases show sharper decay (Huff 38). This implies that high-end grocers, whose offerings differentiate them on quality, curation, and experience rather than price, may draw from considerably wider and more overlapping trade areas than conventional supermarkets. The Huff model thus provides both a theoretical justification and a practical estimation tool for understanding how stores like Erewhon or Bristol Farms compete across Los Angeles's fragmented, auto-oriented urban geography — where store size, brand identity, and neighborhood positioning all shape the probability surface from which each location draws its customer base.

In a more recent approach, Reed et al. examine how specialty grocers respond to different location characteristics including demographic, behavioral, and competitive factors. Reed defines specialty grocery chains as "large regional or national grocery chains in the US offering a more limited assortment of high-end products than what is available in traditional American supermarkets" (Reed 1). Using aspatial and spatial econometric models, Reed identifies several relationships between consumer demographics and specialty grocery store locations. Reed finds that

specialty grocery stores are more likely to locate in urban areas with young, affluent, and well-educated populations than in areas dominated by traditional supermarkets (Reed 5). Additionally, specialty grocers are more likely to be located in trade areas with more pedestrian traffic, including walking and cycling. This connects to their finding that specialty grocers are more likely to be located in areas with more health and fitness establishments, as a concentration of such establishments signals that consumers live healthier lifestyles and have greater demand for the health products characteristic of specialty grocers (Reed 6). Reed suggests that the lifestyles and attributes of consumers are an integral part of location decisions made by specialty grocery chains, as healthy and active consumers signal the possibility of a viable market. Finally, specialty grocers are more likely to be located in trade areas with greater competitive saturation than traditional supermarkets, indicating that high-end grocery stores are not deterred by competition. Rather, competition between specialty grocers can be read as a market signal, indicating that local demand for specialty foods and health-conscious retail is robust enough to sustain multiple entrants (Reed 7).

In “Does Price or Income Affect Organic Choice? Analysis of U.S. Fresh Producer Users,” Smith et al. explore the demand for organic fruits and vegetables using an ordered logit model grounded in random utility maximization. Smith divides households into user and nonuser groups depending on their organic food purchasing habits, further subdividing user households into casual and devoted users depending on their consumption relative to the mean (Smith 734). Smith finds that the probability of a household becoming an organic consumer increases with income, and that the presence of children younger than six further increases this probability, suggesting that parents place greater weight on food safety when young children are

present in the home. Furthermore, urban households are 5.65% more likely to be organic vegetable users, with 4% more likely to be casual users and 1.6% more likely to be devoted users. Notably, households located in the West, where the majority of organic producers and handlers reside, are more likely to purchase organic produce than those located elsewhere in the country (Smith 740). These urban and western household trends point to Los Angeles as a particularly strong organic consumer hub.

Across these bodies of literature, a consistent set of conditions emerges as predictive of both specialty grocery entry and robust consumer demand for premium grocery products: high income, urban density, active and health-oriented lifestyles, proximity to fitness infrastructure, and western regional location. Los Angeles satisfies all of these criteria at the metropolitan level, however, significant variation still exists across neighborhoods and sub-markets. Huff's gravity model provides the method for estimating customer bases in underserved areas; Reed et al. supply the empirical tools for identifying which neighborhood-level conditions signal a viable market gap; Smith et al. quantify the consumer base characteristics that make premium grocery retail financially sustainable, and Hotelling's framework explains how a new entrant can capture a loyal segment without triggering destructive competition with incumbents. Together, these works support the hypothesis that identifiable market gaps for high-end grocery retail exist within Los Angeles.

3. Data

To examine the relationship between neighborhood characteristics and high-end grocery store placement, we constructed a ZIP code–level dataset covering all 270 ZIP codes in Los Angeles County. ZIP codes provide a useful unit of analysis because they are granular enough to capture meaningful variation in income, demographics, and built environment across LA's highly segmented neighborhoods, while still aligning with the geography at which Census data is published. Each observation in our dataset corresponds to one ZIP code and contains a count of high-end grocery stores along with a set of neighborhood-level characteristics drawn from the American Community Survey and each chain's official store locator. The remainder

of this section describes our dependent variable, independent variables, and the sources and limitations of each.

1) Dependent Variable

Store Count:

Erewhon:

<https://erewhon.com/locations>

Whole Foods:

<https://www.wholefoodsmarket.com/stores>

Bristol Farms:

<https://www.bristolfarms.com/stores>

Store Data Construction:

- a. Collect store addresses from official websites
- b. Assign each store to a ZIP code
- c. Count the number of stores per ZIP code

2) Independent variable

Income

<https://censusreporter.org/profiles/16000US0644000-los-angeles-ca/>

House Price

<https://www.zillow.com/>

Rent

<https://censusreporter.org/profiles/16000US0644000-los-angeles-ca/>

(find the dataset: Median gross rent)

Population Density

<https://censusreporter.org/profiles/16000US0644000-los-angeles-ca/>

density = population / area

Median Age

<https://censusreporter.org/profiles/16000US0644000-los-angeles-ca/>

Walkability

https://geohub.lacity.org/datasets/e96fc7d7eb1746cf87ceea7114c6aee9_100/about

Parking

<https://parkinginla.lacity.org/>

Competitor Dummy

Maybe Trader Joe's, Nelson's, Sprouts.

Data source: google map?

Data construction: have competitor=1, not=0

4. Methodology

4.1 Data and Variables

This study uses cross-sectional data to examine the determinants of the number of high-end stores. The dependent variable is the count of high-end stores. Independent variables include median household income, fair market rent for two-bedroom units, population density, median age, and competitor count.

To improve interpretability and reduce skewness, median household income and rental prices are transformed using natural logarithms. Observations with missing values in any of the selected variables are excluded to ensure consistency in

estimation.

4.2 Negative Binomial Model

Given that the dependent variable is a count variable, a Poisson regression model is a natural starting point. However, the Poisson model assumes equality of the conditional mean and variance:

$$E(Y_i | X_i) = \text{Var}(Y_i | X_i)$$

To assess this assumption, we compare the sample mean and variance of the dependent variable and conduct a formal dispersion test. Evidence that the variance exceeds the mean indicates overdispersion, violating the Poisson assumption. To address this issue, we employ a Negative Binomial (NB) regression model, which introduces an additional parameter to allow the variance to exceed the mean, making it more appropriate for overdispersed count data.

4.3 Multicollinearity Diagnostics

Multicollinearity among explanatory variables is assessed using a correlation matrix and Variance Inflation Factors (VIF). Correlation coefficients exceeding 0.8 in absolute value and VIF values above commonly accepted thresholds (e.g., 5 or 10) indicate potential multicollinearity concerns.

4.4 Multicollinearity Treatment

To mitigate multicollinearity, two alternative approaches are considered:

First, a reduced model is estimated by removing highly correlated variables based on theoretical relevance. Second, Principal Component Analysis (PCA) is applied to median household income and rental prices to construct a composite affluence index, defined as the first principal component. This index captures the shared variation between income and rent while reducing dimensionality and collinearity.

4.5 Model Specification

The primary specification is a Negative Binomial regression model of the form:

$$\log(E[Y_i | X_i]) = \beta_0 + \beta_1 \text{Affluence}_i + \beta_2 \text{Density}_i + \beta_3 \text{Age}_i + \beta_4 \text{Competitors}_i$$

4.6 Robust Inference

To account for potential heteroskedasticity, robust (sandwich) standard errors are computed for all models. This ensures that statistical inference remains valid even if model assumptions regarding error variance are violated.

4.7 Model Evaluation

To evaluate model fit, a likelihood ratio (LR) test is conducted to compare the Poisson and Negative Binomial models. A statistically significant result indicates that the Negative Binomial model provides a superior fit to the data.

The final model is selected based on goodness-of-fit, absence of multicollinearity, and interpretability of the estimated coefficients.

5. Results

For our model, first we use a regression model including all variables to predict the relationship between the number of high-end stores and independent variables by collecting previous data. We begin with a baseline regression that includes all independent variables, including income, rent, population density, demographic variables, and competition measures. The results reinforce the profile described above, and clearly show that measures of affluence, particularly those related to income, are most strongly associated with the presence of high-end grocery stores in a ZIP code.

```
> summary(model)

Call:
lm(formula = n_highend_stores ~ log_income + log_rent + population_density +
    median_age + competitor_count, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-0.77098 -0.20267 -0.04451  0.07499  2.70600

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -6.6983807   1.6579749   -4.040 7.04e-05 ***
log_income      0.0017597   0.4092228    0.004 0.996572
log_rent       0.9263069   0.4179608    2.216 0.027540 *
population_density -0.0009796  0.0130382   -0.075 0.940165
median_age     -0.0135466   0.0065827   -2.058 0.040597 *
competitor_count  0.1206764   0.0357945    3.371 0.000862 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3915 on 260 degrees of freedom
Multiple R-squared:  0.2436,    Adjusted R-squared:  0.2291
F-statistic: 16.75 on 5 and 260 DF,  p-value: 2.417e-14
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-6.69838072	2.06697919	-3.2407	0.001348	**
log_income	0.00175972	0.39942011	0.0044	0.996488	
log_rent	0.92630692	0.40479387	2.2883	0.022920	*
population_density	-0.00097963	0.01373136	-0.0713	0.943180	
median_age	-0.01354658	0.00770821	-1.7574	0.080023	.
competitor_count	0.12067638	0.04891870	2.4669	0.014275	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Mean of y: 0.1541353
Variance of y: 0.1987942
> # Detect overdispersion
> if (var_y > mean_y) cat("Variance > mean: suggests overdispersion\n") else cat("No
overdispersion (variance <= mean)\n")
Variance > mean: suggests overdispersion

> print(disptest)
```

Overdispersion test

```
data: poisson_model
z = 1.8526, p-value = 0.03197
alternative hypothesis: true dispersion is greater than 1
sample estimates:
dispersion
1.284889
```

This analysis supports the overall conclusion that high-end grocery retailers are systematically targeting the most affluent neighborhoods in the city where the greatest potential revenue per customer exists. However, since we are including measures of multiple aspects of affluence (income, rent and housing price), there is some concern about multicollinearity. All of the coefficients have the predicted sign but are somewhat imprecise and unreliable. We have run additional tests and considered alternative specifications for these regressions.

As the results above, we can see the min is -0.77098, mean is -0.04451, and the max is 2.70600. The coefficients indicate that log_income, log_rent, and competitor_count are positively correlated with the number of high-end_stores. Severe multicollinearity in the estimates is addressed by assessing and investigating the degree of multicollinearity for each regression. This is done by reporting pairwise correlations and Variance Inflation Factors (VIF) for all regressions. For both regressions, the correlations and VIF values for the key affluence variables, log

income and log rent, were found to be severely above threshold values.

```
> print(cor_mat)
```

	log_income	log_rent	population_density	median_age
log_income	1.0000000	0.7158155	0.8357302	0.6523146
log_rent	0.7158155	1.0000000	0.2520214	0.4756192
population_density	0.8357302	0.2520214	1.0000000	0.5757155
median_age	0.6523146	0.4756192	0.5757155	1.0000000
competitor_count	0.4015342	0.4826401	0.1831350	0.1976736

```
>
> # Identify high correlations (|r| > 0.8)
> high_corr_pairs <- which(abs(cor_mat) > 0.8 & abs(cor_mat) < 1, arr.ind = TRUE)
> if (nrow(high_corr_pairs)==0) cat("No pairs with |cor| > 0.8\n") else print(high_
corr_pairs)
```

	row	col
population_density	3	1
log_income	1	3

```
> vif_model <- lm(n_highend_stores ~ log_income + log_rent + population_density + me
dian_age + competitor_count, data = df)
> vif_values <- vif(vif_model)
> print(vif_values)
```

	log_income	log_rent	population_density	median_age
competitor_count	34.800691	11.624096	18.770137	1.832832
	1.331671			

```
> # Typical rule: VIF > 5 (or 10) indicates multicollinearity concerns
>

> # Option B: combine into an affluence index using PCA (first PC)
> pca_res <- prcomp(df %>% select(log_income, log_rent), scale. = TRUE)
> df$affluence_pc1 <- pca_res$x[,1] # first principal component as index
> # Check VIF after treatment (example using affluence_pc1)
> vif_model_pc <- lm(n_highend_stores ~ affluence_pc1 + population_density + median_
age + competitor_count, data = df)
> print(vif(vif_model_pc))
```

	affluence_pc1	population_density	median_age	competitor_count
	2.327476	1.742157	1.812382	1.331216

Adding inflation, unemployment and/or real income to the benchmark model however creates collinearity problems as these variables are all related to underlying economic conditions and therefore take somewhat duplicate information in the data. In a series of tests we gradually strip away these redundant variables to arrive at a very parsimonious model that is well suited to identify the housing component.

Then we tried to use negative Binomial to process our model, the dependent variable is a count of stores and exhibits, which, in reality, exhibited overdispersion (i.e. variability in the data greater than the mean). Hence a Negative Binomial

regression model was used to estimate the results. The Negative Binomial model offered a better fit for the data in this study than a standard Poisson model.

```
Call:
glm.nb(formula = n_highend_stores ~ affluence_pc1 + population_density +
      median_age + competitor_count, data = df, init.theta = 1497.506069,
      link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    1.56122    1.22621   1.273 0.202944
affluence_pc1    1.43660    0.24828   5.786 7.2e-09 ***
population_density -0.09684    0.02902  -3.337 0.000848 ***
median_age     -0.03592    0.03687  -0.974 0.329972
competitor_count  0.31750    0.15382   2.064 0.039011 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(1497.506) family taken to be 1)

Null deviance: 176.526 on 265 degrees of freedom
Residual deviance: 98.013 on 261 degrees of freedom
AIC: 180.71

Number of Fisher Scoring iterations: 1

              Theta: 1498
            Std. Err.: 17961
Warning while fitting theta: iteration limit reached

2 x log-likelihood: -168.714

> # Robust (sandwich) SEs for NB (coeftest)
> # compute robust vcov (type "HC1")
> vcov_nb_robust <- vcovHC(nb_model, type = "HC1")
> coeftest(nb_model, vcov. = vcov_nb_robust)

z test of coefficients:

              Estimate Std. Error z value Pr(>|z|)
(Intercept)    1.561221    0.920854   1.6954 0.089998 .
affluence_pc1    1.436598    0.200309   7.1719 7.396e-13 ***
population_density -0.096842    0.026234  -3.6914 0.000223 ***
median_age     -0.035918    0.031027  -1.1576 0.247016
competitor_count  0.317496    0.148807   2.1336 0.032875 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Currently, affluence is the leading factor that determines where high-end grocery stores may be likely needed. High-income households are scattered throughout the area, but using data from the Bureau of Labor Statistics that reports the percentage of households in a ZIP code with high incomes, there is a strong correlation between these percentages and the expected number of high-end grocery stores needed to meet the grocery shopping demands of these households. In addition, the competitor variable is positive. This means that high-end grocery stores tend to

cluster in the same neighborhoods as opposed to being spread out across the city. This is consistent with the way that spatial competition is modeled in economic theory, particularly in the famous paper by Hotelling on spatial competition. In that work, firms like the grocery stores in our model locate close to other firms serving the same demand. Other indicators of urbanization, such as population density and walkability, explained only modestly why some affluent suburbs have high-end grocery stores while other affluent suburbs do not; in summary, such stores can be found in both urban and suburban locations, depending on additional local factors.

Due to multicollinearity in the table below (log rent highly correlated with income), the Negative Binomial estimates for income have been re-run with log rent (LRT) dropped from the vector of independent variables. Estimates without LRT generally have much greater precision and are easier to interpret without significant loss of explanatory power.

```
Call:
glm.nb(formula = n_highend_stores ~ log_income + population_density +
  median_age + competitor_count, data = df, init.theta = 982.0835153,
  link = log)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -80.61907   14.01984  -5.750 8.91e-09 ***
log_income      7.61002    1.33709   5.691 1.26e-08 ***
population_density -0.22404    0.04532  -4.944 7.65e-07 ***
median_age     -0.04908    0.03977  -1.234  0.2172
competitor_count  0.34669    0.15374   2.255  0.0241 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(982.0835) family taken to be 1)

Null deviance: 176.51  on 265  degrees of freedom
Residual deviance: 101.77  on 261  degrees of freedom
AIC: 184.48

Number of Fisher Scoring iterations: 1
```

```

      Theta: 982
      Std. Err.: 12544
Warning while fitting theta: alternation limit reached

2 x log-likelihood: -172.483
> coeftest(nb_model_a, vcov. = vcovHC(nb_model_a, type = "HC1"))

z test of coefficients:

              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -80.619075   12.089980  -6.6683 2.589e-11 ***
log_income      7.610018    1.139751   6.6769 2.440e-11 ***
population_density -0.224041  0.040438  -5.5404 3.018e-08 ***
median_age     -0.049079   0.038894  -1.2619  0.20700
competitor_count  0.346687   0.152147   2.2786  0.02269 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The log of income continues to have a highly positive and significant coefficient. It is the best predictor of having a store in the revised specification, and removing the other independent variables from the specification has yielded more stable estimates of the impact of the log of income. The test for the competitor effect remains positively significant at conventional levels, thereby supporting the view of strategic behaviour and precluding the possibility of model misspecification. The model didn't quite hit the numbers I was hoping for, but of these three, the results were by far the most consistent and easiest to understand, which makes it great for figuring out potential market gaps.

After we compare two models between Poisson and Negative Binomial, We assess the adequacy of the specified model using a likelihood ratio test that sets the Poisson model against the Negative Binomial model. The test provides strong evidence against the Poisson model in favour of the Negative Binomial model, showing the presence of overdispersion. We also confirm that a Negative Binomial model is the appropriate primary specification in place of OLS to account for excess

variance in the outcome and to use correct standard errors for proper inference.

```
> print(lr_test)
Likelihood ratio test

Model 1: n_highend_stores ~ affluence_pc1 + population_density + median_age +
  competitor_count
Model 2: n_highend_stores ~ affluence_pc1 + population_density + median_age +
  competitor_count
#Df  LogLik Df  Chisq Pr(>Chisq)
1    6 -84.357
2    5 -84.356 -1  0.0021    0.9639
~ |
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The online appendix also contains comparisons to alternative specifications including a model in which we use a PCA-based affluence index and a model in which we only use log income. While the PCA model has some advantages in terms of capturing more general aspects of socioeconomic status, the log income model has the advantage of being easier to interpret and achieves comparable performance. Starting from the set of possible models, the author discusses the rationale behind choosing the Negative Binomial model, and outlines the several issues with the other models including fit, multicollinearity and interpretation problems. He also discusses the decision to include log income but not log rent. In conclusion, high-end grocery stores tend to cluster in the most affluent areas of Los Angeles, where demand is highest. Rather than shying away from competition, the firms are drawn to the very areas in which other high-end grocers are operating. The most promising vacant space in the market would likely be found in an affluent ZIP code.

Conclusion

This paper looked at where high-end or premium stores like Erewhon, Whole Foods, and Bristol Farms should open their next location in Los Angeles. While we structure it early on identifying “market gaps,” what our model actually does is closer to location selection within the high-end segment by predicting which ZIP codes look most like the neighborhoods where high-end grocers already succeed and ranks them

as options for expansion.

Using a Negative Binomial regression model to handle the overdispersion in our count data, we found that median household income is by far the strongest predictor of where high-end grocery stores show up. After dropping log rent because of its high correlation with income, the log income coefficient stayed positive and significant across every specification, while population density and median age had smaller and less consistent effects. The competitor count variable also came out positive and significant, meaning high-end grocers are usually clustered together or are located not far from each other rather than spread out to different areas. This matches what Hotelling's spatial competition model, an existing premium competitor in a ZIP code, is a positive signal, not a reason to stay out or assume it will affect a high-end store's success in that area.

A chain considering expansion can therefore use fitted values from the model to rank ZIP codes by predicted store count and target the high-ranking ones where they currently have no presence. Limitations include our cross-sectional data, which cannot establish whether grocers cause or follow neighborhood affluence. According to Huff ZIP codes being an imperfect unit since trade areas spill across boundaries, and the lack of per-store revenue data, which means our recommended ZIP codes are demographic matches rather than directly measured demand gaps.

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